

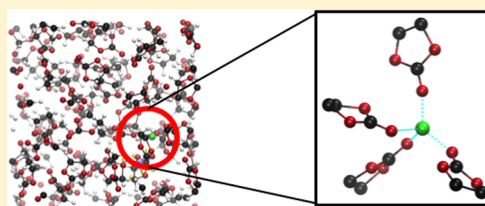
Lithium Ion Solvation and Diffusion in Bulk Organic Electrolytes from First-Principles and Classical Reactive Molecular Dynamics

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ABSTRACT: Lithium-ion battery performance is strongly influenced by the ionic conductivity of the electrolyte, which depends on the speed at which Li ions migrate across the cell and relates to their solvation structure. The choice of solvent can greatly impact both the solvation and diffusivity of Li ions. In this work, we used first-principles molecular dynamics to examine the solvation and diffusion of Li ions in the bulk organic solvents ethylene carbonate (EC), ethyl methyl carbonate (EMC), and a mixture of EC and EMC. We found that Li ions are solvated by either carbonyl or ether oxygen atoms of the solvents and sometimes by the PF_6^- anion. Li^+ prefers a tetrahedrally coordinated first solvation shell regardless of which species are involved, with the specific preferred solvation structure dependent on the organic solvent. In addition, we calculated Li diffusion coefficients in each electrolyte, finding slightly larger diffusivities in the linear carbonate EMC compared to the cyclic carbonate EC. The magnitude of the diffusion coefficient correlates with the strength of Li^+ solvation. Corresponding analysis for the PF_6^- anion shows greater diffusivity associated with a weakly bound, poorly defined first solvation shell. These results can be used to aid in the design of new electrolytes to improve Li-ion battery performance.



1. INTRODUCTION

There is a growing need to replace gasoline and other fossil fuels with environmentally friendly alternative energy sources.¹ However, many of these alternative energy sources such as solar, wind, wave, and geothermal sources require advances in storage technology to become practical. Li-ion batteries are convenient, portable energy storage devices that are currently used to power many handheld consumer devices and electric vehicles. The main components of a Li-ion battery include the anode, cathode, and electrolyte. Carbonaceous materials such as graphite are often used for the anode because of the low cost of carbon and the ease with which Li intercalates into the material. Lithium transition-metal oxides are frequently used for the cathode. The electrolyte provides the medium through which Li ions diffuse between the anode and cathode and usually consists of a lithium salt dissolved in either an organic liquid, an ionic liquid, or a gel polymer.²

The choice of the electrolyte can influence the overall performance of the Li-ion battery.² Many existing batteries use organic liquids to dissolve the Li salt. Among the most common organic liquids used in commercial batteries today are various carbonates, including ethylene carbonate (EC), ethyl methyl carbonate (EMC), and mixtures of these materials. LiPF_6 is a frequently used salt in the electrolyte as it exhibits a high ionic conductivity. A good organic solvent will be able to dissolve a high concentration of salt, resulting in a high dielectric coefficient. A low solvent viscosity facilitates ionic transport.² Typically, cyclic carbonates such as EC have high dielectric

constants but also have high viscosities, whereas linear carbonates such as EMC have lower viscosities but also low dielectric constants. Moreover, some organic liquids such as EC have melting points that are above room temperature, so that they are not liquids over the entire operating temperature range. To resolve these issues and optimize the viscosity, dielectric constant, and melting point of the electrolyte for battery performance, linear and cyclic carbonates are often mixed.² EC is a commonly used liquid for mixed electrolytes because it is known to form a protective layer known as the solid electrolyte interphase (SEI)^{3–9} on graphitic anodes, which prevents excessive electrolyte decomposition and promotes reversible intercalation into and out of the anode.

Experiments have been conducted to examine the effects of the electrolyte composition on Li transport and solvation using different spectroscopic techniques such as Fourier transform infrared (FTIR) spectroscopy,¹⁰ Raman spectroscopy,^{11–14} nuclear magnetic resonance (NMR) spectroscopy,^{11,15,16} electrospray ionization mass spectroscopy,¹⁷ neutron scattering,¹⁸ and X-ray diffraction.¹⁹ Some of these experiments attempted to determine the coordination number around the Li^+ ion using different Li salts, solvents, and concentrations.^{11,12,18} Reported coordination numbers in these works range from ~ 2 to 5. However, there has been relatively little

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experimental characterization of the orientation of solvent molecules around Li^+ . One such study was performed by Cazzanelli et al., who determined a coordination number of ~ 2 in a mixture of EC and propylene carbonate (PC) at different concentrations.¹¹ At high concentrations, they concluded that Li^+ was “sandwiched” between two ring solvent molecules. Recent NMR experiments have also shown that there is a preference for EC to solvate Li^+ over dimethyl carbonate (DMC) in mixed EC/DMC systems and that the carbonyl oxygen atoms are involved in solvation.¹⁶ Separate NMR experiments on transport properties were also carried out, in which experimental diffusion coefficients for Li^+ in different electrolytes were found to fall in the range of $(1\text{--}8) \times 10^{-6} \text{ cm}^2/\text{s}$ at 30°C .²⁰ Mixed EC/EMC systems have Li^+ diffusion coefficients determined to be in the range of $(1.5\text{--}4.5) \times 10^{-6} \text{ cm}^2/\text{s}$, depending on salt concentration.²¹

Theoretical work has also been performed to understand Li transport and solvation in various carbonate electrolytes. To date, many of these simulations were performed using classical force fields.^{22–28} Some of these studies indicated a coordination number of 4, where the Li^+ ion interacts with the carbonyl oxygen atoms of the carbonate.^{22–24,27,28} In mixed carbonate systems such as EC/DMC, it was found that both EC and DMC participate in solvating Li^+ .²² Furthermore, there was a greater affinity for Li^+ to dissociate from its counterion in cyclic carbonates than in linear carbonates.²² However, classical potentials are limited in their transferability and their ability to describe charge-transfer effects. These limitations are not present in first-principles methods, which treat the electrons quantum mechanically. Static quantum calculations using cluster models have been used to study the energetics of different solvation structures,^{29–31} but these studies do not include the effects of the overall liquid environment.

Only recently have first-principles molecular dynamics (FPMD) simulations based on forces from density functional theory been used to study the solvation and transport properties of Li^+ ions in different electrolytes.^{32–34} FPMD can be more predictive than classical force fields because of its parameter-free nature. FPMD has better transferability and can more accurately describe polarizability, charge transfer, and partial charges than classical potentials. It also provides a better basis for future comparative studies of the electrolyte solutions near interfaces and in reactive environments. Previous works using FPMD by both Leung and Budzien³² and Ganesh et al.³³ found Li^+ solvation structures that agreed with previous classical force field simulations and were generally consistent with experiments. In addition, Ganesh et al. calculated diffusion coefficients that were slightly higher than experimental values and previous theoretical results.³³ However, both of these works considered systems with no more than $\sim 300\text{--}400$ atoms and simulation times of $13\text{--}25$ ps, a fact that might have limited the ability to systematically extract quantities of interest. Both also used certain approximations, such as increased hydrogen masses and large time steps, to make the simulations more computationally feasible. Therefore, it remains unclear how accurately these simulations fully describe Li^+ solvation and diffusion in real systems.

In this work, we carried out FPMD simulations to study solvation and diffusion in several experimentally relevant carbonate-based organic Li-ion battery electrolytes. We compared the cyclic carbonate solvent EC to the linear carbonate EMC, as well as an EC/EMC mixture. The typical LiPF_6 salt was used for this work. Solvation and transport

properties for the EC/ LiPF_6 system were previously studied with FPMD,^{32–34} but similar linear carbonate and cyclic/linear carbonate mixtures have been explored only using classical force fields.²³ We analyzed solvation structures of Li^+ to examine how the choice of solvent influences the structure and explore the range of possible solvation structures in each solvent. Furthermore, we examined the interaction of the Li^+ and PF_6^- ions and compared their solvation properties. We calculated the diffusion coefficient in the different electrolytes to understand why Li^+ diffuses faster in one solvent than another and found correlations between solvation and diffusivity. The relation of solvation and diffusivity of PF_6^- was also studied and compared to that of Li^+ . Finally, in addition to performing these studies with larger system sizes and longer time scales than previous FPMD simulations, we further employed the ReaxFF force field³⁵ to quantify the effects of finite size and time scale on the observed Li^+ solvation structures and diffusivity. We anticipate that our findings can be used to design new electrolytes that will improve the cycling rate in batteries by tuning solvation to enhance diffusivity.

2. COMPUTATIONAL DETAILS

We performed first-principles molecular dynamics using density functional theory (DFT) with the projector augmented wave (PAW) method^{36,37} and the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation exchange–correlation functional,^{38,39} as implemented in the Vienna ab initio simulation package (VASP).^{40,41} A 450-eV plane-wave cutoff was used with Brillouin zone sampling restricted to the Γ point. All molecular dynamics simulations were performed in the NVT ensemble using a Nosé–Hoover thermostat,^{42,43} with a Nosé frequency of $\sim 1000 \text{ cm}^{-1}$ corresponding to a period of ~ 32 fs and a time step of 0.5 fs. Each system was equilibrated for 5–7.5 ps at 330 K and then simulated for 30 ps to gather statistics. A temperature of 330 K was used to mimic an intermediate Li-ion battery operating temperature and to ensure that EC was not frozen ($T_{\text{melt}}^{\text{EC}} = 310 \text{ K}$).

Solvation structures were characterized with pair correlation functions, calculated using a bin size of 0.03 Å. Average coordination numbers were computed from the integral of the pair correlation function. We further quantified how tightly the ions were solvated by calculating the average residence time of first-shell solvent molecules by fitting an exponential to the time correlation function

$$P_{\text{solv}}(t) = \langle H(t) \cdot H(0) \rangle \quad (1)$$

where $H(t) = 1$ if a given molecule is within the first solvation shell and 0 otherwise.^{22,23,44,45} The distance cutoff of the first solvation shell was taken from the first minimum in the pair correlation function between the solvated ion and the center of mass of each solvent molecule. A stretched exponential of the form $\exp[-(t/\tau)^\beta]$ gave the best fits, with τ being the residence time and β an adjustable parameter.^{22,23}

Diffusion coefficients for Li^+ and PF_6^- were extracted using two methods: (i) integration of the velocity autocorrelation function (VACF) according to^{46,47}

$$D = \frac{1}{3} \int_0^\infty \frac{1}{N} \sum_{i=1}^N \langle v_i(0) \cdot v_i(t) \rangle dt \quad (2)$$

and (ii) analysis of the mean square displacement (MSD) over time using the Stokes–Einstein relation⁴⁸

$$D = \frac{1}{6} \frac{\langle(\delta r)^2\rangle}{\Delta t} \quad (3)$$

For infinite statistics, using either the MSD or VACF to calculate the diffusion coefficient should produce the exact same answer, as the two quantities are mathematically equivalent. However, for finite statistics, they are not numerically identical,⁴⁹ so we calculated the diffusion coefficient using both methods. The VACF and MSD were both calculated by averaging over multiple trajectory windows spanning the entire trajectory with starting configurations every 50 fs, using various window lengths from 5 to 15 ps in increments of 2.5 ps. Equation 2 was used to calculate D from the VACF for each window length. Likewise, the slope of the linear regime in the MSD was used to calculate D from eq 3 for each window length. In each case, the values for D for each window length were averaged to get a final estimate of the diffusion coefficient. Reported uncertainties reflect the standard deviation of this average.

3. RESULTS AND DISCUSSION

3.1. Li⁺ Solvation in Ethylene Carbonate. We performed first-principles molecular dynamics simulations of a single LiPF₆ molecule dissolved in a periodic box of 63 EC molecules. This system corresponds to 638 total atoms, with a density 1.32 g/cm³ and a Li concentration of 0.23 M. Typical Li concentrations in commercial batteries are 4–5 times larger, but this concentration was used to focus on the solvation of a single Li⁺ ion without the effects of other salt molecules being present. We carried out two independent simulations where the Li⁺ and PF₆[−] ions were initially either associated or dissociated. In Figure 1, we show the Li–P distances over the two

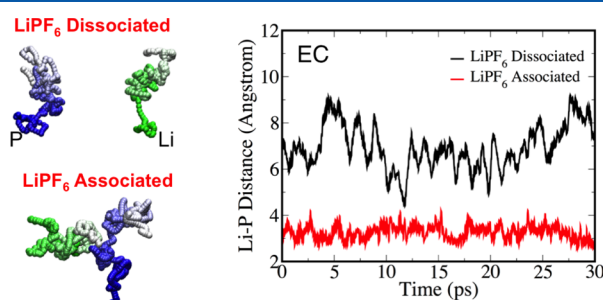


Figure 1. (Left) Trajectories for Li⁺ and P atom of PF₆[−] for the 63EC + 1LiPF₆ system. Color gradients designate time, where darker colors are earlier in time. (Right) Li–P distance as a function of time for trajectories that are initially dissociated (black) and associated (red).

trajectories, which indicate that LiPF₆ remains either associated or dissociated for the entire simulation. In addition, we display the trajectories of the Li⁺ ion and the P from the PF₆[−] ion where the color gradients (dark to light) indicate time. We found that Li⁺ and PF₆[−] follow similar trajectories even when dissociated, with a separation of ~ 5 – 8 Å, providing evidence of correlated motion.

The pair correlation functions between the Li ion and either the carbonyl oxygen atoms from EC, designated O_C, or the ether oxygen atoms, designated O_E, are shown in Figure 2. We found two different solvation structures: one in which PF₆[−] stays apart from Li⁺ (Figure 2a) and another in which they remain close (Figure 2b). For the case where they remain associated, we observed an average coordination number of

~ 2.5 for Li–O_C and ~ 0.5 for Li–O_E in the first solvation shell. In this case, PF₆ also occupies a site in the first solvation shell with a total coordination number of 4. We label this solvation structure 3carbonylPF6. In the case where Li and PF₆ are dissociated, we calculated a coordination number of 4 for Li–O_C. We denote this solvation structure 4carbonyl. Representative snapshots of the 3carbonylPF6 and 4carbonyl solvation structures are shown in Figure 2. The peak of the Li–O_C pair correlation function at approximately ~ 1.9 Å agrees with the results of previous classical^{23–25} and FPMD^{33,34} simulations, which range between 1.7 and 2.0 Å. Our total coordination number of 4 also agrees with previous theoretical work^{22,24,27,28,32–34} and experiments.¹⁸ We compared the thermodynamic stabilities of these two structures by computing the average relative energies over the trajectories, which are also indicated in Figure 2. We found the 4carbonyl solvation structure to be favorable by ~ 0.2 eV. We also examined the orientation of the solvent molecules around Li⁺ in each case by tracking the O_C–Li–O_C angle. Figure 3a shows histograms of the O_C–Li–O_C angles for both the 3carbonylPF6 and 4carbonyl solvation structures. Both are peaked at $\sim 110^\circ$, indicating a preference for a tetrahedral arrangement in both cases. We also found that the carbonyl group of the EC molecule tends to point toward the Li⁺ ion, with a Li–O_C–C_C angle of $\sim 140^\circ$, as shown in Figure 3b. Previous FPMD simulations by Ganesh et al.³³ also determined this angle to be 140° , which is consistent with the experimental value of 138° ¹⁸ but lower than the classical force field value²³ of 150° . The tetrahedral pattern we observed for 4carbonyl agrees with previously calculated solvation structures for Li⁺ in EC.^{22,24,27,28,32–34} The near- 110° angle for 3carbonylPF6 suggests that, regardless of the composition of the first solvation shell, Li⁺ prefers to be solvated in a tetrahedral fashion.

3.2. Li⁺ Solvation in Ethyl Methyl Carbonate. Unlike EC, ethyl methyl carbonate (EMC) has a lower viscosity and freezes at a much lower temperature, but it also has a lower dielectric constant. We examined how the solvation structures and ionic motion differed when a linear carbonate such as EMC was used as the solvent. The simulation system consisted of 42 EMC molecules and 1 LiPF₆ molecule (638 total atoms) with a density of 1.01 g/cm³ and a concentration of 0.22 M. Again, we ran simulations with the LiPF₆ initially either associated or dissociated. We found that the initially dissociated LiPF₆ reassociated within 15 ps during the simulation, as shown in the Li–P distance plot in Figure 4. On the other hand, when LiPF₆ started associated, it remained associated throughout the simulation. Upon examination of the trajectories, we found that when Li⁺ and PF₆[−] were initially dissociated, the PF₆[−] ion moved toward and found the Li⁺ ion. Overall, PF₆[−] was observed to migrate farther than Li⁺. This is likely due to the Li⁺ ion being more tightly solvated than the PF₆[−] ion, as discussed later.

We observed three different solvation structures during the simulations. The pair correlation functions of Li–O_C and Li–O_E for each of these solvation structures along with representative snapshots and relative energies are shown in Figure 5. We found that the lowest-energy structure is one where the Li⁺ and PF₆[−] ions are associated (Figure 5a). This is a significant difference compared to EC. The total coordination number of Li⁺ in this preferred structure is 4, consisting of three carbonyl oxygens from three EMC molecules and the nearby PF₆[−] ion. We designate this solvation structure as 3carbonyl-

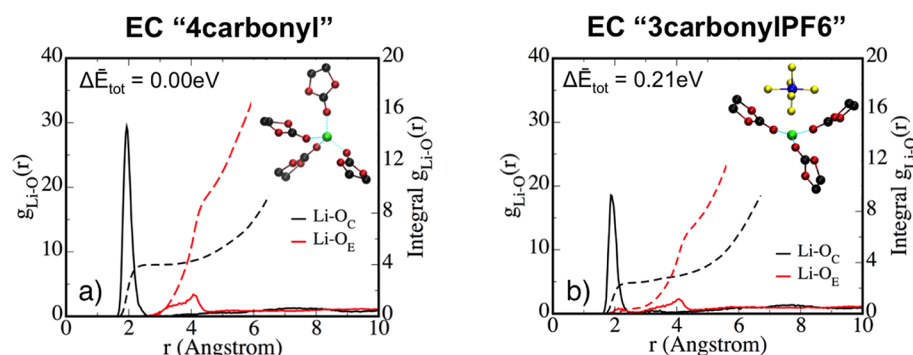


Figure 2. Li–O_C and Li–O_E pair correlation functions (solid lines) and their integrals (dashed lines) for (a) 4carbonyl and (b) 3carbonylPF6 solvation structures of EC. Snapshots of the solvation structures and average energies relative to the lowest-energy structure are shown in the insets.

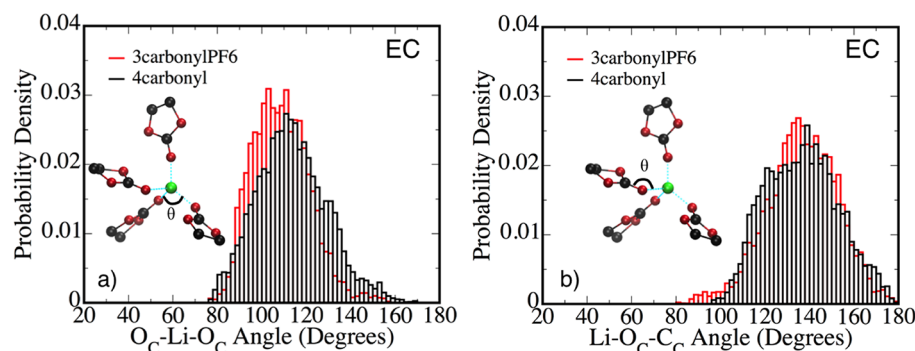


Figure 3. Histograms of representative (a) O_C–Li–O_C and (b) Li–O_C–C_C angles (denoted in the insets) for the 3carbonylPF6 and 4carbonyl solvation structures of EC along the trajectory.

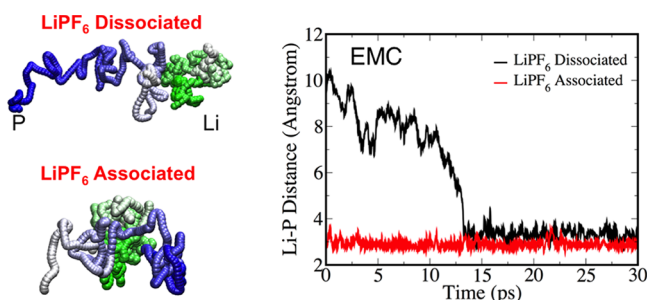


Figure 4. (Left) Trajectories for Li⁺ and P atom of PF₆[−] for the 42EMC + 1LiPF₆ system. Color gradients designate time, where darker colors are earlier in time. (Right) Li–P distance as a function of time for trajectories that are initially dissociated (black) and associated (red).

yIPF6. We note that a similar structure was also observed with EC, but it was not the lowest-energy configuration. There is another solvation structure in EMC that is 4-fold-coordinated and consists of four EMC molecules oriented such that Li⁺ is solvated by four carbonyl oxygen atoms and one ether oxygen atom. In this case, the Li⁺ and PF₆[−] ions remain apart; we denote this structure as 3carbonylether (Figure 5b). The orientations of these two 4-fold-coordinated solvation structures are analyzed in Figure 6a, where we plot the O_C–Li–O_C bond angle for the 3carbonylPF6 and 3carbonylether cases. For both of these cases, there is a peak near 110°, similar to EC, indicating a preferred tetrahedral arrangement of the solvent molecules regardless of whether EMC or PF₆[−] is solvating Li⁺. We also note a preference for the carbonyl oxygen to point toward Li⁺, from the Li–O_C–C_C angle of ~150° shown in Figure 6b. Similar conclusions were drawn for EC, but

the Li–O_C–C_C angle was slightly larger for EMC than for EC, which is also consistent with trends seen by Borodin and Smith²³ using classical force fields. Unlike for EC, however, we also found a nontetrahedral structure for EMC in which the Li⁺ ion is solvated by four total ether oxygen atoms belonging to two EMC molecules (two ether oxygen atoms per EMC) and PF₆[−], in a square pyramidal-like fashion. We label this structure 4etherPF6 (Figure 5c). This structure is not energetically preferred, being more than 0.4 eV higher in energy than 3carbonylPF6. For EMC, a corresponding 4carbonyl solvation structure, which is favored by EC, was not observed in either of the trajectories. Steric issues likely prohibit this structure for EMC, as the length of the molecule makes it unfavorable to have four EMC molecules around Li⁺. With PF₆[−] similar in size to EC, the 3carbonylPF6 structure is preferred for EMC, instead of the 4carbonyl structure as for EC.

3.3. Li⁺ Solvation in a 3:7 Mixture of EC/EMC. We also examined the effect of mixing different organic solvents for the electrolyte. We studied a mixture of EC and EMC in a 3:7 ratio, mimicking previous experiments.⁷ The simulation system consisted of 15 EC molecules, 35 EMC molecules, and 1 LiPF₆ molecule (683 total atoms) with a density of 1.165 g/cm³ and a concentration of 0.23 M. As in the previous studies, we began the simulations with the LiPF₆ either associated or dissociated. In Figure 7, we plot the Li–P distance for both cases and show the trajectories of the two ions. One can see that the ions remain associated or dissociated for the duration of the simulation. In addition, based on the trajectories, one can see that the Li and P in both the associated and dissociated cases follow very similar paths, indicating that the ions behave more like in EC than in EMC, with a large degree of correlated motion even when the ions are separated. This similarity to EC

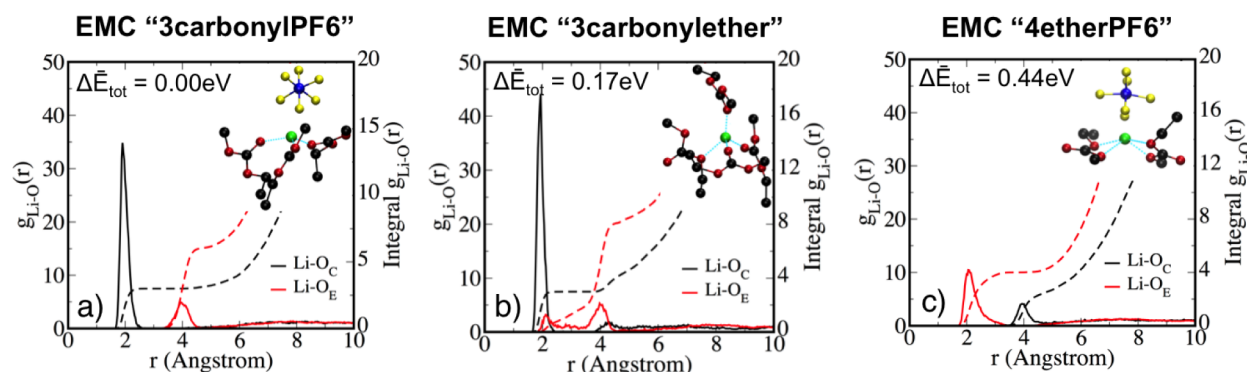


Figure 5. Li–O_C and Li–O_E pair correlation functions (solid lines) and their integrals (dashed lines) for (a) 3carbonylPF6, (b) 3carbonylether, and (c) 4etherPF6 solvation structure of EMC. Snapshots of the solvation structures and average energies relative to the lowest-energy structure are shown in the insets.

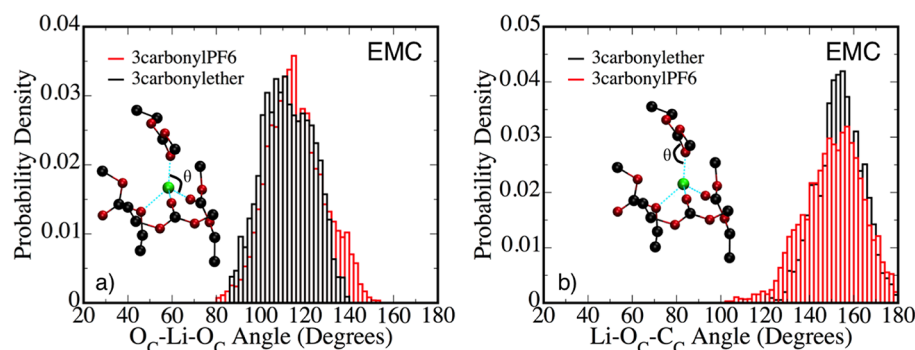


Figure 6. Histograms of representative (a) O_C–Li–O_C and (b) Li–O_C–C_C angles (denoted in the insets) for 3carbonylPF6 and 3carbonylether solvation structures of EMC along the trajectory.

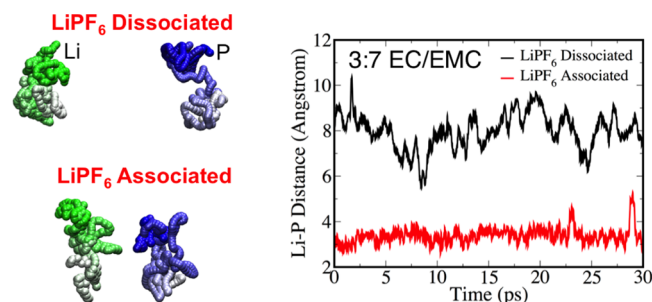


Figure 7. (Left) Trajectories for Li⁺ and P atom of PF₆[−] for the 1SEC + 35EMC + 1LiPF₆ system. Color gradients designate time, where darker colors are earlier in time. (Right) Li–P distance as a function of time for trajectories that are initially dissociated (black) and associated (red).

occurs even with the mixture containing only 30% EC. Therefore, adding only a small percentage of EC to the system results in a dramatic change in the ion motion with respect to the original system.

From these trajectories, we found three different solvation structures: two structures in which the Li⁺ and PF₆[−] ions stay apart and one structure in which they remain together. Snapshots of the solvation structures, their pair correlation functions for Li–O_C and Li–O_E, and their relative energies are shown in Figure 8. For the two solvation structures with LiPF₆ dissociated, one shows a coordination number of 4 for Li–O_C, indicating that four carbonyl oxygen atoms solvate Li⁺; the other has a coordination number of ~3 for Li–O_C and ~1 for Li–O_E, where three carbonyl oxygen atoms and one ether

oxygen atom solvate Li⁺. We denote these solvation structures as 4carbonyl (Figure 8a) and 3carbonylether (Figure 8b), respectively. For the one structure in which LiPF₆ is associated, there is a coordination number of 2 for Li–O_C and ~1 for Li–O_E, indicating that two carbonyl oxygen atoms and one ether oxygen atom solvate Li⁺ along with the PF₆[−] ion. We label this structure 2carbonyletherPF6 (Figure 8c). The 4carbonyl and 3carbonylether solvation structures are nearly energetically equivalent, with only a 0.04-eV difference between the two, whereas the 2carbonyletherPF6 is more than 0.3 eV higher in energy. In this 3:7 EC/EMC mixture, Li⁺ prefers to be separated from PF₆[−], similarly to pure EC. In addition, we observed that one EMC molecule in the 3carbonylether solvation structure is replaced by an EC molecule during one of the trajectories, leaving two EC molecules and two EMC molecules involved in the first solvation shell and forming the 4carbonyl solvation structure. This result indicates a strong preference for EC molecules to solvate Li⁺, considering the small fraction of EC in the system.

Recent ¹⁷O NMR experiments performed on mixed EC/DMC systems also showed a strong preference for EC to solvate Li⁺ as opposed to a linear carbonate such as DMC (or EMC).¹⁶ This is consistent with our observation that one EMC molecule was replaced by an EC molecule in the first solvation shell during the course of our simulation. In addition, classical simulations also observed equal amounts of cyclic carbonate, EC, and linear carbonate, DMC, when the ions were dissociated.²³ Therefore, our results are consistent with experiments and classical simulations and provide strong evidence that Li⁺ prefers to be solvated by EC when present

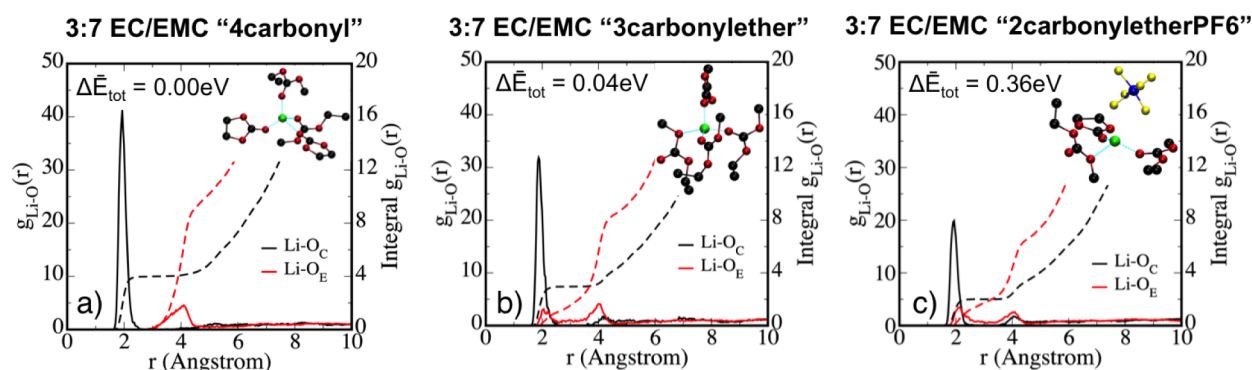


Figure 8. Li–O_C and Li–O_E pair correlation functions (solid lines) and their integrals (dashed lines) for (a) 4carbonyl, (b) 3carbonylether, and (c) 2carbonyletherPF6 solvation structures of mixed EC/EMC. Snapshots of the solvation structures and average energies relative to the lowest-energy structure are shown.

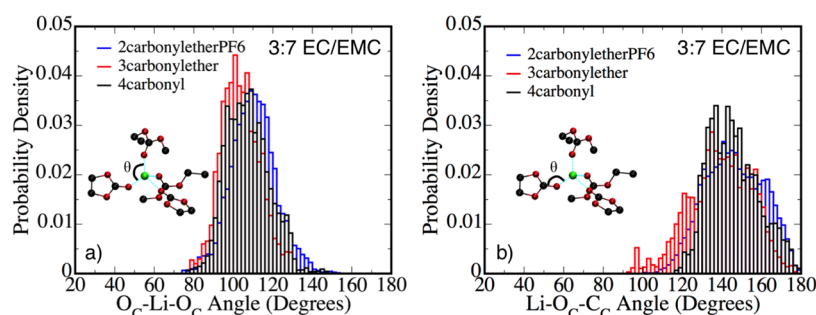


Figure 9. Histograms of representative (a) O_C–Li–O_C and (b) Li–O_C–C_C angles (denoted in the insets) for the 4carbonyl, 2carbonyletherPF6, and 3carbonylether solvation structures of mixed EC/EMC.

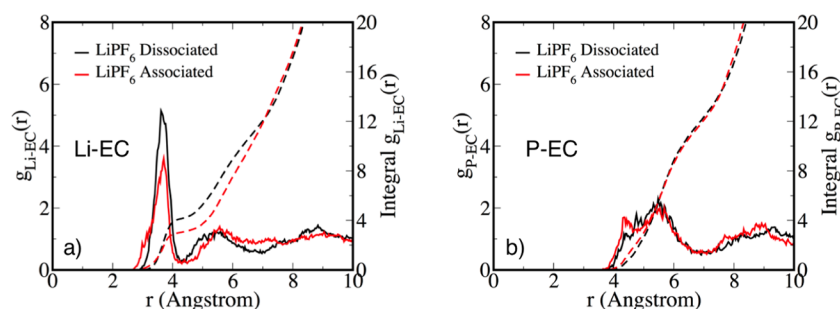


Figure 10. (a) Li–EC and (b) P–EC pair correlation functions (solid lines) and their integrals (dashed lines) for LiPF₆ in EC, with the salt molecule initially either associated (red) or dissociated (black).

in these systems. As described above, this tendency likely is related to steric effects and the fact that EC and PF₆ have similar sizes. We also examined the orientation of the solvent molecules for all three cases. In Figure 9a, we show histograms of the O_C–Li–O_C angle for each solvation structure. Again, we found the peak of the histogram for all three cases near 110°, with Li⁺ preferring a tetrahedral solvation structure. In Figure 9b, we plot the Li–O_C–C_C angle and note that the carbonyl oxygen atoms prefer to point toward Li⁺ with an angle of ~140°, similar to the previous cases (and particularly similar to pure EC).

Overall, our results show that even small variations in the organic solvent can dramatically change the preferred solvation structure, although Li always prefers to be coordinated tetrahedrally in its first solvation shell regardless of which species are around it.

3.4. Solvation Structure of PF₆[−]. Thus far, we have primarily examined the solvation of the Li ion. We now turn

our attention to the solvation structure of the PF₆[−] anion and compare it to that of Li⁺. There was relatively little discussion about the solvation structure of the counterion in previous FPMD and classical simulations in the dissociative limit. The EC system is used here for illustration, although EMC and the mixture show similar phenomena. In Figure 10, we show the pair correlation functions of the center of mass of the EC molecules and either Li (Figure 10a) or P (Figure 10b), for both trajectories where the ions were initially either associated or dissociated. The Li–EC pair correlation functions are well-structured with a sharp first-shell peak, indicating that Li⁺ has a well-defined solvation structure with a clear coordination number of ~4 in the first solvation shell, as already discussed in detail. Conversely, the P–EC pair correlation functions are very broad, suggesting that many EC (solvent) molecules dynamically rotate in and out of the first solvation shell on a short time scale. To quantify the strength of solvation, we computed the average residence time of the first-shell solvent

molecules around each of the ions. For PF_6^- , the residence time of EC was 43–90 ps, and that of EMC was 24–29 ps. (The ranges correspond to variations for trajectories with LiPF_6 either associated or dissociated.) For Li^+ , the residence times were well beyond the length of the simulations for both EC and EMC, with the first solvation shells showing little solvent exchange during the trajectories. Whereas previous FPMD simulations^{33,34} for EC reported no solvent exchange during their trajectories, we observed occasional solvent exchange that preserved the tetrahedral solvation structure. Thus, we determined that PF_6^- is much more weakly solvated, with a poorly structured solvation shell and very short solvent molecule residence times, than Li^+ . Also, we found that EC solvates PF_6^- somewhat better than EMC, based on the relative residence times.

We have already shown in Figures 1, 4, and 7 that PF_6^- appears to move farther than Li^+ , despite the much heavier mass of the anion. The increased diffusivity of PF_6^- compared to Li^+ is related to the respective solvation structures and is discussed in detail in the following section.

3.5. Li^+ and PF_6^- Transport Properties. The transport properties of Li^+ and PF_6^- in each of the electrolytes were studied by comparing their diffusion coefficients. Table 1

Table 1. Li^+ Diffusion Coefficients in Each Electrolyte Calculated from the Slope of the Mean-Square Displacement (MSD) and the Integral of the Velocity Autocorrelation Function (VACF)^a

electrolyte composition	Li^+ diffusion coefficient ($\times 10^{-6} \text{ cm}^2/\text{s}$)	
	MSD	VACF
63EC + 1 LiPF_6	5.2 ± 0.8	7.9 ± 1.3
42EMC + 1 LiPF_6	9.6 ± 1.6	10.1 ± 2.1
1SEC + 35EMC + 1 LiPF_6	2.6 ± 1.3	5.1 ± 1.1

^aFor each electrolyte, the most stable solvation configuration(s) was (were) considered.

compares the Li^+ diffusion coefficients for each electrolyte, calculated using both the MSD and VACF methods (see section 2 for details). We focused on the cases with solvation in the preferred configurations, which generally correspond to LiPF_6 being dissociated, as discussed above. Trends in the diffusion coefficients were generally consistent across the different systems regardless of whether MSD or VACF was used for the calculations. Statistical errors of the calculated values were on the order of $(1\text{--}2) \times 10^{-6} \text{ cm}^2/\text{s}$ as shown in Table 1 or as much as 50%. Improving these errors is nontrivial, as it would require running many (>20) independent simulations with one Li ion or a single simulation with many more Li ions (and a correspondingly larger system size).

Values for EC were approximately a factor of 2 different than previous FPMD results,^{33,34} which calculated the diffusion coefficient to be $\sim 1 \times 10^{-5} \text{ cm}^2/\text{s}$ at temperatures between 310 and 450 K, likely due to the large uncertainty associated with the short DFT runs. The highest Li diffusion was seen in EMC. This is in agreement with experiments²⁰ and classical simulations,²³ which found faster diffusion in linear carbonates compared to cyclic carbonates. Surprisingly, the mixed electrolyte showed slower Li^+ diffusion than pure EC, whereas it would be expected to fall between EC and EMC. We suspect the reason why the diffusion coefficient of the mixture does not fall between EC and EMC is related to statistical error from the rather short DFT simulation and the variation of solvation

structure during the trajectory. However, we note that the similar solvation structures of Li^+ in EC and the EC/EMC mixture result in similar diffusivities in these cases, which are distinct from that in pure EMC. In fact, the first solvation shell of Li^+ in the EC/EMC mixture can contain up to two EC molecules, as discussed above, so it is expected that the value of the diffusion coefficient would be closer to that of pure EC.

Furthermore, analysis of the range of trajectories shows that slower diffusion tends to occur for cases in which the Li^+ solvation structure is more energetically preferred. A similar observation was noted when the coordination number was greater than 4 (i.e., the nontetrahedral EMC case in Figure 5c). These results reveal that the magnitude of the diffusion coefficient is strongly dependent on how tightly solvated the Li^+ is by its solvent molecules. When the Li^+ is more tightly solvated, the diffusion coefficient is smaller than when it is weakly solvated. We conclude that EC solvates Li^+ better than EMC, as indicated by the lower diffusion coefficient.

Table 2 reports the diffusion coefficients for PF_6^- , calculated by tracking the P atom. Overall, the values are larger than for

Table 2. PF_6^- Diffusion Coefficients in Each Electrolyte Calculated from the Slope of the Mean-Square Displacement (MSD) and the Integral of the Velocity Autocorrelation Function (VACF)^a

electrolyte composition	PF_6^- diffusion coefficient ($\times 10^{-6} \text{ cm}^2/\text{s}$)	
	MSD	VACF
63EC + 1 LiPF_6	7.1 ± 0.9	9.2 ± 1.0
42EMC + 1 LiPF_6	30.8 ± 8.8	28.6 ± 5.7
1SEC + 35EMC + 1 LiPF_6	5.7 ± 2.4	9.5 ± 1.4

^aFor each electrolyte, the most stable solvation configuration(s) was (were) considered.

Li^+ , consistent with the weaker solvation structure discussed above. We also note the higher diffusivity in EMC compared to EC, which is due to the weaker solvation by EMC as evidenced by the shorter residence time of solvent molecules in the first solvation shell.

3.6. Finite-Size and Time-Scale Effects. Molecular dynamics simulations based on traditional Kohn–Sham density functional theory implementations are limited to moderate system sizes on the order of hundreds of atoms and time scales of tens of picoseconds. To gauge finite-size and time-scale effects on the solvation structures and diffusion coefficients that we calculated using DFT, we used the ReaxFF reactive force field^{35,50,51} as implemented in the LAMMPS^{52,53} software package to run much longer molecular dynamics trajectories of 1 ns with system sizes up to ~ 6400 atoms.

First, we assessed the quality of the ReaxFF force field to reproduce the results of DFT, using the EC electrolyte as a test case. These simulations were performed under NVT conditions using a Nosé–Hoover chain thermostat with three Nosé–Hoover chains and a time step of 0.25 fs. The Nosé frequency was set to $\sim 1333 \text{ cm}^{-1}$, corresponding to a period of ~ 25 fs. A system of 630 EC and 10 LiPF_6 molecules was equilibrated for 125 ps at 330 K and then simulated for 1 ns. This test system contained the same concentration of LiPF_6 as the DFT simulation but with 10 times more Li ions to gather better statistics.

In Figure 11a, we show the Li–O pair correlation function and its integral for both the ReaxFF trajectory and the DFT trajectory. We found that both methods predict the same

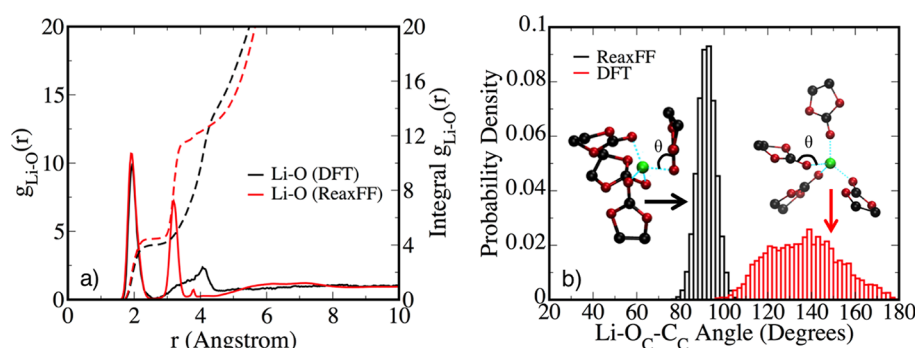


Figure 11. (a) Li–O pair correlation functions (solid lines) and their integrals (dashed lines) for dissociated LiPF_6 in EC, comparing DFT and ReaxFF trajectories. (b) Histogram indicating the distribution of the Li-O-C-C angle for ReaxFF and DFT trajectories. Insets show representative snapshots of the ReaxFF and DFT Li^+ solvation structures.

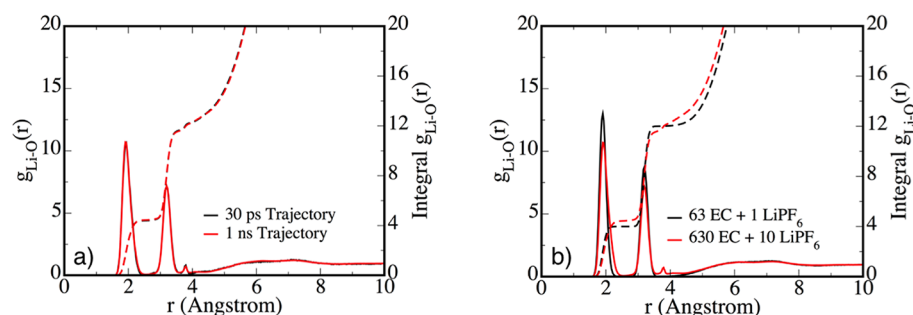


Figure 12. (a) Comparison of Li–O pair correlation functions and their integrals for a 30-ps trajectory and a 1-ns trajectory. (b) Comparison of Li–O pair correlation functions and their integrals for the 63EC + 1 LiPF_6 system size and the 630 EC + 10 LiPF_6 system size, where the concentrations of the two systems are fixed.

coordination number of ~ 4 carbonyl oxygen atoms for the first solvation shell. Also, the O-C-Li-O-C angle distributions are nearly identical, with a peak at $\sim 110^\circ$ indicating a tetrahedral arrangement, as shown, for example, in Figure 3. However, the second peak in the Li–O pair correlation function (associated with the ether oxygen atoms) is slightly different, with DFT predicting a broader peak centered farther away than ReaxFF. This second peak is much sharper and more structured with ReaxFF. One can understand this difference more deeply by examining the Li-O-C-C angle. Figure 11b shows that DFT exhibits a broad distribution of angles centered at $\sim 140^\circ$, whereas ReaxFF predicts a narrow distribution of angles around $\sim 90^\circ$. Representative snapshots of these solvation structures from ReaxFF and DFT are displayed in the insets of Figure 11b, which illustrate how the 90° angle from ReaxFF results in the ether oxygen atoms being closer to Li^+ than in the DFT simulations. In addition, the second Li–O peak in the pair correlation function from ReaxFF is much sharper than that from DFT, because this “bent” solvation structure is more rigid, presumably from additional interactions between Li and O_E . In the DFT simulations, the carbonyl dipoles of the EC molecules point toward the Li^+ but also exhibit more rotational fluctuations, giving rise to the broadening of the second peak in the Li–O pair correlation function and of the Li-O-C-C angular distribution.

ReaxFF also predicts occasional EC dimerization over the course of the trajectory, where the carbonyl carbon of one EC molecule interacts with the carbonyl oxygen of another EC molecule. This dimerization is inconsistent with DFT and the chemical inertness of the EC liquid but only occurs for less than 5% of the molecules over 1 ns. We further compared the diffusion coefficients for both Li^+ and PF_6^- and found that the

ReaxFF values are within ~ 40 – 50% of the DFT values. Thus, we do note differences between DFT and ReaxFF for these systems, but the applicability of ReaxFF to study finite-size and time-effects appears valid.

To determine the effects of time scale on the solvation structure, we calculated the Li–O pair correlation function and its integral using a small 30-ps segment of the 630EC + 10 LiPF_6 1-ns ReaxFF trajectory. In Figure 12a, these results are compared to the pair correlation function obtained when the entire 1-ns trajectory was used. We see that peak locations, intensities, and coordination numbers are very similar, indicating that the time scale used for the MD simulations does not have a significant effect on the solvation structure. Similarly, to determine the effects of finite size on the solvation structure, we ran a 63EC + 1 LiPF_6 MD simulation with ReaxFF for 1 ns under NVT conditions at 330 K. The Li–O pair correlation function and integral for this 63EC + 1 LiPF_6 system are compared to those calculated for the 630EC + 10 LiPF_6 system in Figure 12b. One can see that, although the peak intensities vary slightly, the peak locations and the integral of the pair correlation function are comparable. These results were confirmed with several additional system sizes as well. Based on these observations, we do not expect the solvation structures calculated using DFT to change significantly when going to longer time scales or larger system sizes at fixed concentrations.

We also examined the effects of time scale and finite size on the diffusion coefficients for both Li^+ and PF_6^- in pure EC using ReaxFF. We compared the diffusion coefficient for Li^+ and PF_6^- , calculated using the slope of the mean square displacement, from a 30-ps segment of the 630EC + 10 LiPF_6 trajectory to that from the entire 1-ns trajectory. We found that using only 30 ps of the trajectory resulted in a difference of

~46% for Li^+ and 38% for PF_6^- as opposed to using the whole 1-ns trajectory in the calculation. Based on these results, we expect that our DFT-calculated diffusion coefficients could vary by up to 50% upon running longer simulations. Classical simulations were also used previously to study size effects and found a 10% difference in the diffusion coefficient between large (480 solvent molecules) and small (240 solvent molecules) box sizes.²³ However, more Li atoms were included in these simulations, which increased the statistical sampling and resulted in correspondingly smaller finite-size effects. We further compared the diffusion coefficients calculated from 1-ns trajectories of systems containing either 63EC + 1LiPF₆ or 630EC + 10LiPF₆, to judge the effects of finite size without changing the statistical sampling. The diffusion coefficients between the smaller and larger systems differed by ~6% for Li^+ but by up to ~48% for PF_6^- . Although the discrepancy for PF_6^- is fairly large, it is comparable to the approximately 50% uncertainty that we found for the time-scale effects. Tests on EMC and the EC/EMC mixture showed results similar to those for EC.

To summarize, we found that the Li^+ solvation structures do not change significantly when assessing finite-size and time-scale effects, although there are some differences between the ReaxFF and DFT solvation structures. Furthermore, uncertainties up to ~50% in the diffusion coefficients are expected for simulations run at short time scales and smaller length scales. This uncertainty is still small enough to allow us to draw qualitative and semiquantitative conclusions as above, given the relative statistical errors in our computed values. However, it will be important in the future to use DFT at still larger length and time scales to further reduce uncertainties in solvation structures and diffusion coefficients.

4. CONCLUSIONS

We found multiple possible solvation structures of Li^+ in each of the electrolytes studied here, including ethyl carbonate (EC), ethyl methyl carbonate (EMC), and an EC/EMC mixture. Whereas previous literature on EC and LiPF₆ focused on solvation by four carbonyl oxygen atoms, we observed solvation structures that also include PF_6^- in the first solvation shell. We found that the preferred solvation structure is strongly dependent on the solvent. For EC, Li^+ prefers to be solvated by four carbonyl oxygen atoms and to remain dissociated from PF_6^- , whereas for EMC, it shows some preference for only three carbonyl oxygen atoms and to remain close to the PF_6^- ion. The 3:7 EC/EMC mixture shows a preference for Li^+ to remain dissociated from PF_6^- , but has two energetically similar solvation structures where Li^+ is solvated either by four carbonyl oxygen atoms or by three carbonyl oxygen atoms and one ether oxygen atom. In all cases, Li^+ prefers to be solvated in a tetrahedral arrangement, although this does not rule out the possibility of a slightly higher-energy nontetrahedral structure forming, as seen in the EMC 4etherPF6 case. Comparisons of solvation structures for Li^+ and PF_6^- reveal that Li^+ is more strongly solvated, associated with lower mobility in the electrolyte. Calculations of residence times of first-shell solvent molecules show that there is a slight preference for PF_6^- to be solvated by EC over EMC, although both show weak solvation of PF_6^- .

Calculated diffusion coefficients quantify the ionic motion in each electrolyte and relate to the solvation structures. We found that the largest Li^+ diffusion coefficient occurs in EMC. This is consistent with the measured viscosities of the two organic

solvents, and the diffusivity values obtained agree well with experimental values. The magnitude of the diffusion coefficient is largely influenced by how tightly the ion is solvated, for both Li^+ and PF_6^- . A more tightly bound solvation structure, such as for Li^+ in EC, leads to slower diffusion of the solvated ion. Comparison of Li^+ and PF_6^- diffusion coefficients indicates that PF_6^- diffuses faster than Li^+ in all of the electrolytes examined here, even though it is the heavier species. This can be attributed to the fact that the solvent molecules interact more strongly with Li^+ than with PF_6^- .

Furthermore, we quantified finite-size and time-scale effects on the solvation structures and diffusion coefficients using ReaxFF. We found that solvation structures do not change significantly for larger system sizes and longer time scales than used here with DFT, but there are some structural differences between ReaxFF and DFT simulations. Absolute diffusion coefficients are more affected by size and time-scale effects, as uncertainties can be as large as 50%, but relative values remain consistent.

Our work has shown that a more tightly bound solvation structure leads to slower diffusion and a weakly bound solvation structure leads to faster diffusion. To improve the mobility of Li ions in solution, our results suggest that Li^+ must have weak interactions with the organic solvent used in the electrolyte. This is a valuable insight that can be used to improve the cycling rate of Li-ion batteries and potentially lead to the design of new electrolytes for better overall battery performance.

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Notes

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